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Tool unit, press for holding such a tool unit and method for a ready-to-operate arrangement of the tool unit in the holder of the press

Abstract

A tool unit **20** and a press **21**, particularly an embossing press. The tool unit **20** is simply and quickly arranged in or removed from a holder **23** of the press **21** in order to carry out a quick tool change with short downtime of press **21**. For this purpose the tool unit **20** comprises a first tool **45** and a second tool **46** that are arranged along a tool axis A on a tool carrier **31**. By means of a tool guide device **47**, the second tool **46** is movably supported in a guided manner relative to the tool carrier **31** and the first tool **45**, independent from guide devices of press **21**. Therefore, the tool unit **20** can be prepared and set up during the ongoing operation of press **21**. If desired or necessary, the prepared tool unit **20** is exchanged against the presently used tool unit **20**.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This is a continuation-in-part application of pending international patent application PCT/EP2020/059730 filed Apr. 6, 2020 and claiming the priority of German Patent Application No. 10 2019 109 700.2 filed Apr. 12, 2019. The said international patent application PCT/EP2020/059730 and said German Patent Application No. 10 2019 109 700.2 are both incorporated herein by reference in their entireties as though fully set forth.

BACKGROUND OF THE INVENTION

(1) The invention refers to a tool unit for a press, a press with a holder for such a tool unit as well as a method in order to arrange the tool unit in a condition ready for operation in the press. The method also comprises the exchange of an actually used tool unit with another tool unit.

(2) Particularly in fast operating presses, e.g. embossing presses for embossing coins, medals, round blanks or other plate-shaped blanks, the exchange and set-up of tools for forming a blank, particularly a round blank, is very time consuming. Especially during minting of coins the embossing tools have to be very precisely orientated with regard to one another due to the filigree structures that have to be minted in a coin or medal. The set-up of a press prior to operation is quite time consuming. If during the operation wear occurs on the tools, it can be necessary to stop the press to exchange and re-orientate the tools and to subsequently resume operation. So far used embossing presses have therefore relatively long downtime durations.

(3) A minting press is, for example, known from DE 26 28 855 C2. It comprises a holder for a female die (first tool) as well as a ram supporting a male die (second tool). The holder for the first tool consists of semi-circular bowls that form a hollow space for the first tool in the clamped condition. The two tools can be exchanged separately and have to be orientated relative to one another in the context of start-up of the press, such that optimum embossing results are obtained.

(4) DE 10 2015 010 489 A1 describes a forging press having a tool carrier. The tool carrier has a sliding surface. For exchange one or multiple tools can be slidably supported in a guiding manner on this sliding surface.

(5) Starting from the known prior art, it can be considered as object of the present invention to provide a tool unit for a press, particularly an embossing press, that can be inserted or removed with short downtimes of the press.

(6) This object is solved by means of a tool unit, a press, as well as a method having the features of the claims.

SUMMARY OF THE INVENTION

(7) The inventive tool unit for a press has a tool carrier that is configured to be exchangeably connected with the press. Particularly the tool carrier is configured to be arranged, positioned and/or orientated in a respective holder on a press frame of the press and to be releasably attached to the press frame. The tool unit forms a unit that can be uniformly handled in order to arrange it in a press or remove it from a press.

(8) The tool unit has in addition a first tool as well as a second tool. The two tools are arranged in alignment with a tool axis on the tool carrier. The two tools are so-to-speak coaxially arranged to the tool axis A.

(9) The tool unit comprises a tool guide device by means of which the second tool is movably supported in a guided manner on the tool carrier along the tool axis. In doing so, the second tool can be moved in a stroke direction toward the first tool and opposite the stroke direction in a back-stroke direction away from the first tool. The two tools are thus orientated relative to one another and guided by means of the tool unit itself. If the tool unit is inserted in the press, the second tool can be coupled with a ram of the press by means of a respective coupling device such that the movement of the ram can be transmitted on the second tool and thus a stroke in stroke direction or a back-stroke in back-stroke direction can be initiated.

(10) In that the two tools are already movably guided and orientated relative to one another already in the tool unit, the insertion and removal of the tools in/from the press is remarkably simplified. The tool unit is handled as module or unit in a body, such that a cumbersome alignment of the two tools relative to one another in the press is omitted. Different to the presses known so far, particularly embossing presses, the second tool does not use the guide device of the ram, but the tool unit has a separate own tool guide device for the second tool.

(11) Preferably the tools of the tool unit are embossing tools for an embossing press, in order to emboss, for example, coins, medals or the like from a plate or disc-shaped blank or a round blank. The blank or round blank can be a one-part blank or a multiple-part blank consisting of a core and one or multiple rings arranged concentrically around the core. The outer contour of the round blank or a round blank part can be, for example, circular or polygonal or can have another substantially arbitrary shape.

(12) It is advantageous, if the tool guide device is configured to support transverse movements of the second tool transverse to the tool axis and/or tilting movements of the second tool relative to the tool axis. Particularly the tool guide device can be configured such that the second tool can only be moved in a guided manner along the tool axis in one single degree of freedom and movements in all other degrees of freedom—apart from a technically necessary clearance—are supported.

(13) In one embodiment the tool guide device can have a guide column that is movably supported in an assigned guide bushing parallel to the tool axis. Thereby, for example, the first tool can be movably coupled with the at least one guide column, for example, it can be immovably arranged relative to the guide column. The at least one guide bushing can be provided on the tool carrier.

(14) It is advantageous, if the second tool comprises a second tool coupling surface by means of which the second tool abuts on a ram surface of a ram of a press at least during its movement in stroke direction. Preferably the tool coupling surface and the ram surface are planar orthogonal to the tool axis. The second tool and the ram are configured in the area of the tool coupling surface and the ram surface abutting against each other, such that a tilting movement of the ram relative to the tool axis and/or transverse movements of the ram orthogonal to the tool axis are not transmitted on the second tool.

(15) In an embodiment of the tool unit the first tool is movable between a forming position and an ejecting position. In the ejecting position a formed round blank can be ejected from a ring after the forming or embossing. In the ejecting position the first tool is supported in stroke direction, particularly immovably and can thereby be supported either on the tool carrier or on the press.

(16) As an alternative in another embodiment it is also possible that the first tool is immovably arranged relative to the tool carrier.

(17) A transport part having a sliding surface can be preferably provided on the tool carrier. The transport part having the sliding surface is configured to allow the sliding transport of round blanks in the tool unit or out of the tool unit. Thereby the round blanks can be positioned between the two tools in order to form or emboss and/or punch them subsequently and to cart them off over the sliding surface after embossing and/or punching.

(18) It is in addition advantageous, if a movement guide device is provided on the tool carrier. The movement guide device is configured to cooperate with a frame guide device on a press frame of the press. By means of the movement guide device and the frame guide device the tool unit can be

moved into the holder of the press or out of the holder of the press.

(19) In a preferred embodiment at least one positioning recess is provided on the tool carrier of the tool unit. The at least one positioning recess is configured to cooperate with a positioning body of a positioning device of the press. Due to insertion of the at least one positioning body in the at least one positioning recess, the tool carrier and thus the tool unit can be positioned or orientated relative to a press frame of the press. Originating from its open end the at least one positioning recess comprises a tapering cross-section and can thus be particularly conical. The at least one positioning projection is preferably configured complementarily to the at least one positioning recess.

(20) An inventive press for forming a round blank comprises a holder for an exchangeable tool unit, particularly for a tool unit according to any of the embodiments, as they have been described above. The press has a positioning device having at least one positioning body. The positioning body can be moved in an alignment position in that it projects at least partly in the respectively assigned positioning recess on the tool unit. Due to the cooperation between the at least one positioning body and the at least one positioning recess, positioning and/or orientation of the tool unit relative to the press can occur. In addition, the press comprises a clamping device that is configured to clamp the tool unit in the holder. The clamping device is particularly configured such that during creation of the clamping force no form-fit is created by means of the clamping device, but a force-fit or friction-fit connection between the clamping device and the tool unit is maintained. It is particularly advantageous, if the press is configured to move the at least one positioning body out of the alignment position after clamping of the tool unit, such that no contact between the at least one positioning body and the at least one positioning recess exists. In another embodiment the clamping device can be alternatively or additionally configured such that a form-fit is created during creation of the clamping force.

(21) In a preferred embodiment the press has a coupling device. The coupling device is configured to couple the second tool with the ram of the press, such that the movement of the ram can be transferred to the second tool in order to move it in stroke direction or in back-stroke direction.

(22) It is preferred, if the coupling device is configured to push or pull the second tool in back-stroke direction with a coupling force toward the ram. A stop surface can be provided on the press that can be preferably positioned or adjusted against which the second tool is pressed or pulled by means of the coupling force, if it takes a rest position. In the rest position the second tool has a maximum distance to the first tool, if the first tool is in the forming position. In the rest position the second tool is not lifted from the stop surface by the ram. The ram is located in the range of its return point distant from the first tool in back-stroke direction that can also be denoted as lower return point.

(23) It is advantageous, if the coupling device is configured to allow transverse movements of the ram transverse to the tool axis and/or tilting movements of the ram relative to the tool axis. In doing so, it is avoided that such transverse movements or tilting movements of the ram are transferred to the second tool. This avoids a clamping of the second tool that is guided by means of the tool guide device on the tool unit independent from the guidance of the ram.

(24) For a ready-to-operate arrangement of a tool unit in the holder of a press it is particularly proceeded as follows:

(25) First, the tool unit is inserted in the holder of the press. The movement guide device on the tool holder and the frame guide device on the press frame can be used for this purpose, for example, and the tool unit can be moved transverse to the tool axis in the holder on the press frame. Subsequently, the tool unit is positioned and/or orientated relative to the press frame. For this the at least one positioning body is moved into the respectively assigned at least one positioning recess on the tool unit by means of the positioning device such that the at least one positioning body gets in contact with the tool unit in the at least one positioning recess, such that the position and/or orientation of the tool unit relative to the press frame is defined. Subsequently the tool unit is clamped on the press frame by means of a clamping device. After clamping the at least one

positioning body can be retracted from the respectively assigned positioning recess again, such that no contact exists between the at least one positioning body and the tool unit.

(26) It is in addition advantageous, if a movement coupling between the first tool and the ram of the press is established by means of a coupling device of the press, particularly in that the second tool is urged with a coupling force in back-stroke direction against the ram.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Advantageous embodiments of the invention are derived from the dependent claims, the description and the drawings. In the following, preferred embodiments of the invention are explained in detail based on the attached drawings. The drawings show:

- (2) FIG. 1 a schematic block diagram-like illustration of a tool unit arranged in a holder of a press during positioning and/or orientating;
- (3) FIG. 2 the block diagram of FIG. 1 after alignment in a condition ready for operation during the supply of a round blank in the tool unit;
- (4) FIG. 3 the embodiment according to FIGS. 1 and 2 during embossing of the supplied round blank;
- (5) FIG. 4 the embodiment according to FIGS. 1-3 during the ejection of the round blank after embossing;
- (6) FIG. 5 the embodiment according to FIGS. 1-4 in which the round blank has been positioned incorrectly prior to embossing;
- (7) FIG. 6 a perspective illustration of an embodiment of a tool unit and a press prior to insertion of the tool unit in the holder of the press;
- (8) FIG. 7 the embodiment of the tool unit and the press of FIG. 6 in another perspective view;
- (9) FIG. 8 the embodiment of the tool unit and the press according to FIGS. 6 and 7 after the tool unit has been inserted in the holder of the press;
- (10) FIG. 9 a perspective view of the embodiment of the tool unit according to FIGS. 6-8;
- (11) FIGS. 10-13 a sectional illustration through the tool unit and the press according to FIGS. 6-9 respectively, wherein the tool unit is arranged in the holder of the press; and,
- (12) FIGS. 14 and 15 a perspective sectional illustration of an embodiment of the tool unit according to FIGS. 6-13 respectively.

DETAILED DESCRIPTION OF THE INVENTION

(13) In FIGS. 6 and 7 a perspective illustration of an embodiment of a tool unit **20** as well as a press **21** is respectively illustrated, wherein the press **21** is only shown partly. The press **21** is configured as embossing press, particularly minting press, to form one piece or multi-piece round blanks in coins, medals or the like.

(14) The press **21** has a press frame **22** with a holder **23** that is configured for holding a tool unit **20**. In the embodiment a frame guide device **24** is provided in the holder **23** that cooperates with a movement guide device **25** provided on the tool unit **20**, such that the tool unit **20** can be moved orthogonal to a stroke direction H and a back-stroke direction R in the holder **23** on the press frame **22**. FIG. 8 illustrates the tool unit **20** completely inserted in the holder **23**.

(15) In the preferred embodiment the frame guide device **24** comprises two guide projections **26** arranged with distance opposite to one another that form a guide rail in each case. The guide projections **26** or guide rails extend in an insertion direction orthogonal to a stroke direction H and a back-stroke direction R. The stroke direction H and the back-stroke direction R are orientated opposite to one another and define the direction in which a ram **27** of the press **21** and/or a tool of the tool unit **20** coupled with the ram **27** can be moved for forming a round blank. The movement of the ram **27** is initiated by means of a ram drive **28** of the press **21**.

(16) The tool unit **20** has a tool carrier **31** on which the movement guide device **25** is configured or provided. In the embodiment the movement guide device **25** comprises a groove-like movement recess **32** respectively that is limited by two groove flanks **33**, **34**, as shown in FIG. **1**, that are arranged with distance opposite to one another. In the embodiment the stroke direction H and the back-stroke direction R are substantially vertically aligned such that one groove flank forms a top groove flank **33** while the groove flank arranged vertically below is a bottom groove flank **34**, if the tool unit **20** is orientated in a position in which it can be inserted in the holder **23**. During insertion of the tool unit **20** in the holder **23** of the press **21** an assigned guide projection **26** projects in the movement recess **32** respectively. Thereby the top groove flank **33** is placed on top of the respectively assigned guide projection **26** in the embodiment.

(17) In FIGS. **1-5** the decisive parts of the tool unit **20** as well as elements of the press **21** are illustrated in a kind of block diagram respectively. FIGS. **14** and **15** show a perspective sectional illustration through a specific embodiment of the tool unit **20** according to FIG. **6-13** respectively.

(18) In the block diagram illustrations according to FIGS. **1-5**, the tool unit **20** is arranged in the holder **23** of the press **21**. Complete insertion of the tool unit **20** in the holder **23** is, for example, detected by means of a sensor device **38**. The sensor device **38** is illustrated in FIG. **3**, for example, and for the sake of clarity omitted in FIGS. **1**, **2** and **4**. In the embodiment the sensor device **38** comprises a light transmitter **39**, for example a laser or the like, as well as a light receiver **40**. A through-hole **41** is provided on the tool carrier **31** of the tool unit **20** that clears the light path between the light transmitter **39** and the light receiver **40**, if the tool unit **20** is completely and correctly arranged in the holder **23**. If the tool unit **20** is not in the correct position in the holder **23**, the tool carrier **31** or at least a part thereof blocks the light path between the light transmitter **39** and the light receiver **40**, such that dependent on the signal at the light receiver **40** a correct or incorrect arrangement of the tool unit **20** in the holder **23** can be recognized.

(19) Instead of the light transmitter **39** and the light receiver **40**, also a different sensor technology can be used such as a proximity switch or a contact switch that can be operated electrically and/or magnetically.

(20) The tool unit **20** comprises a first tool **45** as well as a second tool **46** that are arranged in alignment along a common tool axis A. According to the example, the first tool **45** is arranged vertically above the second tool **46**, if the tool unit **20** is arranged in the holder **23**. The tool unit **20** also comprises a tool guide device **47** by means of which the second tool **46** can be moved in stroke direction H toward the first tool **45** and opposite in the back-stroke direction R away from the first tool **45**. Preferably the tool guide device **47** is configured such that tilting movements of the second tool **46** relative to the tool axis A and/or transverse movements of the second tool **46** radially or orthogonal to the tool axis A are supported on the tool carrier **31**—apart from a technically necessary clearance. The tool guide device **47** is thus configured as linear guide for movement of the second tool **46** along the tool axis A.

(21) In the illustrated preferred embodiment the tool guide device **47** comprises two guide columns **48** that extend parallel to one another with distance from the tool axis A in stroke direction H or back-stroke direction R. Each guide column **48** is movably guided in an assigned guide bushing **49** in stroke direction H and back-stroke direction R. The guide bushings **49** are arranged on the tool carrier **31** or in the tool carrier **31**. For this purpose cavities or through-holes can be provided in the tool carrier **31**, for example, in which one of the guide bushings **49** is arranged in each case. The guide bushings **49** can be configured as friction guide bushings or also as roller guide bushings for the guide columns **48**.

(22) As schematically apparent from FIGS. **1-5** and in an embodiment of the tool unit **20**, particularly from FIG. **15**, the two guide columns **48** are arranged immovably relative to the second tool **46**. For this a transverse support **50** and e.g. a plate-shaped transverse support **50** is provided in the embodiment, on which the two guide columns **48** are attached. In addition, the second tool **46** is attached on the transverse support **50**. The transverse support **50** can be an integrally formed body.

(23) The two tools **45**, **46** of the tool unit **20** are preferably multi-part tools, which is particularly apparent from FIG. **15**. The first tool **45** has a first pressure piece **51** on which a first embossing stamp **52** is attached. The second tool **46** comprises a second pressure piece **53** on which a second embossing stamp **54** is attached correspondingly. In modification thereto, however, it would also be possible to configure the first tool **45** and the second tool **46** in a one-piece integral manner respectively.

(24) In addition, a transport part **57** is attached on the tool carrier **31** that is according to the example configured in a plate-shaped manner. The transport part **57** comprises on its side facing toward the first tool **45** a sliding surface **58** that extends substantially orthogonal to the tool axis A in one plane. The sliding surface **58** is configured to allow a sliding transport of round blanks **59** to be formed or to be embossed. For this purpose a respective transport device, not shown in the drawings, can be provided in order to transport round blanks **59** in the tool unit **20** and after forming or embossing of the respective round blank **59** out of the tool unit **20**. For example, a rotary disc, not shown in the drawings, can be used as transport device that is stepwise rotatable around a rotation axis and comprises multiple transport pockets for holding one round blank **59** respectively with distance to its rotation axis. In each rotational position of the rotary disc one transport pocket is aligned with the tool axis A of the tool unit **20**. The round blank **59** present in one of the transport pockets slides on the sliding surface **58** during rotation of the rotary disc in a position in which it is located between the two tools **45**, **46** and can be formed or embossed by means of a stroke of the second tool **46** relative to the first tool **45**.

(25) Subsequently, the embossed round blank **59** can be ejected back into the transport pocket, e.g. by means of an ejector device **62**. The ejector device **62** can comprise, for example, an ejector **63** that is movable in stroke direction H and back-stroke direction R and on which the first tool **45** is supported, if the tool unit **20** is arranged in the holder **23** (FIG. **10**). By means of the ejector device **62** and particularly the ejector **63**, the first tool **45** can be moved between a forming position I (FIGS. **1-3** and **5**) and an ejecting position II (FIG. **4**).

(26) The ejector device **62** is required, if an embossed round blank **59** does not readily fall by itself through the weight force back into the transport pocket. In the embodiment a ring **64** is provided adjacent to the first tool **45**, if it is in the forming position I, wherein the ring **64** is arranged on the tool carrier **31** and is supported on the tool carrier **31** in a spring-elastic manner. During forming or embossing of the round blank, the round blank **59** is located inside ring **64** (FIG. **3**). The ring **64** limits the radial expansion of the round blank **59** during forming or embossing. After forming or embossing the round blank **59** can be held or clamped in the ring **64** in a force-fit manner. By means of a relative movement between ring **64** and first tool **45** the embossed round blank **59** can be ejected from ring **64** and conveyed back in the transport pocket.

(27) In the embodiment the first tool **45** is moved from the forming position I into the ejecting position II for this purpose, such that the first tool **45** projects at least partly into ring **64** and ejects the formed or embossed round blank **59** as schematically shown in FIG. **4**. For this the ejector **63**, on which the first tool **45** is supported, is moved in back-stroke direction R. Subsequently the embossed round blank **59** can be discharged by means of the transport device and according to the example by means of the rotary disc from the tool unit **20** and a new round blank **59** to be embossed can be supplied that is located in the adjacent transport pocket of the rotary disc, not shown in the drawings.

(28) For movement coupling of the second tool **46** with the ram **27** of the press **21**, the press **21** comprises a coupling device **68**. The coupling device **68** is configured to apply a coupling force to the second tool **46** that urges the second tool **46** in back-stroke direction R toward the ram **27** of the press **21**. In the embodiment in the coupled condition the coupling device **68** applies a tensile force on the second tool **46** in back-stroke direction R (FIGS. **2-5**). In a rest position of the second tool **46** (FIG. **2**) a tool coupling surface **69** of the second tool **46** is in abutment against a stop surface **70** of press **21** or press frame **22**. In the embodiment the stop surface **70** is configured as ring surface

that is arranged coaxially to the ram 27. The tool coupling surface 69 is facing the ram surface 71 of ram 27. In the rest position illustrated in FIG. 2 the ram surface 71 can be arranged with distance to the tool coupling surface 69.

(29) According to the example, the tool coupling surface 69 is orientated in a plane orthogonal to the tool axis A. As illustrated by way of example, the ram surface 71 extends substantially orthogonal to the stroke direction H or back-stroke direction R. The ram surface 71 can at least deviate in one or more positions of the ram 27 from the ideal parallel orientation relative to the tool coupling surface 69 or can move during the movement in stroke direction H or back-stroke direction R, also orthogonal to the stroke direction H or back-stroke direction R. Due to the coupling device 68, it is avoided that such an undesired ram orientation or ram transverse movement is transmitted on the second tool 46 in that the coupling device 68 does not establish a movement coupling in these degrees of freedom between the ram 27 and the second tool 46. Preferably only the movement coupling in stroke direction H or back-stroke direction R is established by means of the coupling device 68.

(30) According to the example, the coupling device 68 has a coupling part 72 that is connected by means of an appropriate lever arrangement or another connection with an actuator unit 73. The coupling part 72 is urged and thus biased in back-stroke direction R relative to the press frame 22 by means of a biasing device 74, e.g. a spring arrangement. The actuator unit 73 is configured to position or move the coupling part 72 against the force of the biasing device 74 in stroke direction H. Thereby the actuator unit 73 can serve as stop in order to inhibit a movement of the coupling part 72 in back-stroke direction R due to the biasing force of the biasing device 74 (FIG. 1). As soon as the actuator unit 73 is moved in a condition or position in which the coupling part 72 can be freely moved by the biasing device 74 in back-stroke direction R, the coupling part 72 can be brought in abutment with a counter coupling part 75 that is coupled with the second tool 46. The counter coupling part 75 is immovably arranged relative to the second tool 46 according to the example and can be provided on the transverse support 50 or can be formed by a part of the transverse support 50.

(31) As particularly apparent from FIGS. 10 and 12 in an embodiment, a coupling surface 76 is provided on the coupling part 72 that particularly corresponds to a ball shell surface section. A counter coupling surface 77 is provided on the counter coupling part 75 that is complimentary to the coupling surface 76 and according to the example also corresponds to a ball shell surface section. In the embodiment the coupling surface 76 is convex, while the counter coupling surface 77 is concave. Also a vice versa arrangement is possible as an alternative to the illustrated embodiment. The surfaces are in sliding movable abutment, if a coupling between the ram 27 and the second tool 46 is established (FIGS. 2-5). The coupling surface 76 and the counter coupling surface 77 can also have other contours than the form having concave or convex ball shell surface sections illustrated in FIGS. 10 and 12.

(32) Because the coupling surface 76 and the counter coupling surface 77 are in a slidably movable abutment against one another, it is achieved that the coupling device 68 allows tilting movements of the ram 27 of the press 21 relative to the tool axis A and/or transverse movements orthogonal to the tool axis A and does not or only in a negligible manner transfer such movements on the second tool 46.

(33) The coupling device 68 can comprise multiple coupling parts 72 that cooperate with respectively one counter coupling part 75 of the tool unit 20. On the coupling part 72 a biasing force can be applied by a common or by respectively one biasing device 74 and can comprise a common or respectively one actuator unit 73. In the embodiment according to FIGS. 6-15 the press 21 comprises two coupling parts 72 having a biasing device 74 and an actuator unit 73 in each case.

(34) By means of the coupling device 68 of the press 21 it is achieved that the second tool 46 carries out a movement in stroke direction H, if ram 27 moves in stroke direction H and the ram surface 71 abuts against the tool coupling surface 69 (FIG. 3). In doing so, a round blank 59 can be

lifted from the sliding surface **58** and moved into ring **64** in order to form or emboss the round blank **59** inside the ring **64** between the tools **45**, **46**.

(35) The tool coupling surface **69** abuts against the stop surface **70** in the rest position of the second tool **46**. It is possible that the second tool **46** carries out only a portion of the stroke movement of the ram **27**. The stroke of the ram **27** of the press **21** in stroke direction H can be longer than the movement in stroke direction H carried out by the second tool **46**.

(36) In addition, the press **21** has a positioning device **82** having at least one and according to the example two positioning bodies **83**. The positioning bodies **83** are movable in stroke direction H and back-stroke direction R by means of a positioning drive **84** in each case. The positioning drive **84** can be realized by a cylinder.

(37) The positioning bodies **83** taper in back-stroke direction R toward their free ends and are conically formed according to the example. Each positioning body **83** is assigned to a positioning recess **85** provided on the tool unit **20** and according to the example on the tool carrier **31**. The positioning recesses **85** are open in stroke direction H in the embodiment. They taper inwardly in back-stroke direction R and are particularly conically formed analog to the respectively assigned positioning body **83**.

(38) By means of the positioning drives **84** the positioning bodies **83** can be moved in an alignment position III (FIG. 1) in which the positioning bodies **83** project into the respective assigned positioning recess **85** and abut inside the positioning recess **85** against the tool carrier **31**. The tool unit **20** is orientated relative to the press frame **22**, because the positioning bodies **83** as well as the positioning recesses **85** are formed in a manner tapering and particularly conically tapering in back-stroke direction R. The frame guide device **24** and the movement guide device **25** are provided with a respective clearance. The holder **23** of the press frame **22** is therefore configured, such that a relative movement between the tool unit **20** and the press frame **22** is possible.

(39) In addition, the press **21** has a clamping device **90**. The clamping device **90** is configured to effect a clamping force between the tool carrier **31** and the press frame **22** and according to the example the movement guide device **25** and the frame guide device **24**. In doing so, the tool unit **20** is clampingly fixed in the holder **23** of the press frame **22**. In the embodiment the clamping device **90** has multiple and according to the example four clamping bodies **91** for this purpose that are movable by means of a respectively assigned clamping drive **92** in a clamping position IV (FIGS. 2-5). In the embodiment the clamping bodies **91** are moved by means of a respectively assigned clamping drive **92** in back-stroke direction R or stroke direction H. The clamping drive **92** can be realized, for example, by a cylinder and particularly a hydraulic cylinder. In the clamping position IV the clamping body **91** is pressed against the tool carrier **31** in back-stroke direction R by means of the clamping drive **92**. In the embodiment thereby the top groove flanks **33** are pressed with the clamping force against the respectively adjacent guide projection **26**. A force-fit attachment of the tool unit **20** in the holder **23** is achieved. The clamping bodies **91** abut only in a force-fit manner against the tool carrier **31** and do not establish a form-fit connection. In doing so, it is guaranteed that the relative position and the relative orientation between the tool unit **20** and the press frame **22** adjusted by means of the positioning device **82** is not affected by the clamping device **90**.

(40) FIG. 11 illustrates an embodiment of the clamping device **90** in which the clamping drive **92** comprises respectively one spring stack and particularly one disc spring stack that is supported on the piston **94** of a cylinder **95**. The assigned clamping body **91** is arranged on the piston **94** of the respective clamping drive **92** and projects out of the cylinder **95**. The disc spring stack **93** urges the piston **94** together with the clamping body **91** in back-stroke direction R and against the tool carrier **31** in the clamping position IV in order to create the clamping force. A chamber in the cylinder **95** can be subject to a fluid pressure, e.g. a hydraulic pressure, in order to move the piston **94** together with the clamping body **91** in stroke direction H out of the clamping position IV against the force of the disc spring stack **93** and the clamping of the tool unit **20** in the holder **23** can be disabled. Due to the creation of the clamping force with the spring stack and particularly a disc spring stack

93, a clamping effect is guaranteed independent from externally available energy and particularly a fluid pressure.

(41) The arrangement of the tool unit **20** in the press **21** is executed as follows according to the example:

(42) Prior to inserting the tool unit **20** in the holder **23**, the positioning device **82**, the clamping device **90** and the coupling device **68** are moved in a position in which the insertion or slide movement of the tool unit **20** into the holder **23** is possible. Subsequently, the tool unit **20** is moved into the holder **23**, e.g. by using the movement guide device **25** and the frame guide device **24**, until the tool unit **20** is completely and correctly arranged in the holder **23**. This is detected by means of the sensor device **38** (FIG. 3). As soon as the tool unit **20** is correctly inserted in the holder **23**, a positioning, orientating, attaching and coupling of the tool unit **20** and the press **21** is carried out. This can be done either entirely automatically after correct arrangement of the tool unit **20** in the holder **23**, which is recognized by means of the sensor device **38**, or can be done by an operating person after initiating a respective command.

(43) First, the positioning device **82** is activated and the positioning bodies **83** are moved in their respective alignment position III. In doing so, the positioning bodies **83** project in the respectively assigned positioning recess **85** and align the tool unit **20** relative to the press frame **22**. This condition is schematically illustrated in FIG. 1.

(44) Subsequently, the clamping device **90** and the coupling device **68** are concurrently or subsequently activated. Preferably the clamping bodies **91** of the clamping device **90** are first moved in their respective clamping position IV and thus a clamping attachment of the tool unit **20** in the holder **23** of the press frame **22** is effected. After this attachment the positioning bodies **83** of the positioning device **82** are moved out of their alignment position III, such that a contact between the positioning bodies **83** and the tool unit **20** or the tool carrier **31** does no longer exist (FIG. 2). In addition, the coupling device **68** or the actuator unit **73** is operated such that the second tool **46** is urged by means of a biasing force of the biasing device **74** in the rest position against the stop surface **70** (FIG. 2). Now the press **21** is ready for operation.

(45) Now a round blank **59** can be supplied between the two tools **45**, **46** for embossing. The ram **27** is moved in stroke direction H by means of the ram drive **28**, such that the ram surface **71** gets into contact with the tool coupling surface **69**. In case of a continued stroke of the ram **27** in stroke direction H, the second tool **46** is moved together with it in stroke direction H. Thereby the supplied round blank **59** is moved by means of the second tool **46** along the tool axis A toward the first tool **45** that is in the forming position I and is formed between the two tools **45**, **46** (FIG. 3). The first tool **45** is supported in the forming position I at the ejector device **62** and does not move in stroke direction H. The round blank **59** is located inside the ring **64** during forming that limits the radial expansion of the round blank **59**.

(46) After forming the round blank **59** has an outer diameter that corresponds to the inner diameter of ring **64**. The formed or embossed round blank **59** is held in a force-fit manner in the ring **64**, if the second tool **46** moves away from the first tool **45** due to the back-stroke movement of the ram **27** in back-stroke direction R until it takes its rest position in which the tool coupling surface **69** abuts against the stop surface **70** (FIG. 4). In order to discharge the formed or embossed round blank **59** from ring **64**, the ejector device **62** moves the first tool **45**—at least temporarily concurrently or subsequent to the back-stroke movement of the second tool **46**—in back-stroke direction R into ring **64** and presses the formed or embossed round blank **59** out of ring **64** (FIG. 4). Thereby the first tool **45** is in the ejector position II. Subsequently the first tool **45** is moved back again in forming position I while the formed or embossed round blank **59** is cart off the tool unit **20** and a new round blank to be formed or to be embossed is supplied. This forming or embossing process is repeated in a cyclic manner.

(47) In the embodiment the sensor device **38** has an additional function for monitoring the forming or embossing that is schematically illustrated in FIG. 5. If a round blank **59**, that is not arranged in

alignment along the tool axis A with the interior of ring **64**, is lifted by means of the second tool **46**, the round blank **59** does not reach the interior of ring **64**. Rather the ring **64** is displaced due to the contact with the round blank **59** against its spring support into the light path and interrupts the light path between the light transmitter **39** and the light receiver **40**. Thereby an error during forming or embossing of a round blank **59** can be recognized.

(48) The invention refers to tool unit **20** and a press **21**, particularly an embossing press. The tool unit **20** can be simply and quickly arranged in or removed from a holder **23** of the press **21** in order to carry out a quick tool change with short downtime of press **21**. For this purpose the tool unit **20** comprises a first tool **45** and a second tool **46** that are arranged along a tool axis A on a tool carrier **31**. By means of a tool guide device **47**, the second tool **46** is movably supported in a guided manner relative to the tool carrier **31** and relative to the first tool **45**, independent from guide devices of press **21**. Therefore, the tool unit **20** can be prepared and set up during the ongoing operation of press **21**. If desired or necessary, the prepared tool unit **20** is exchanged against the presently used tool unit **20**.

LIST OF REFERENCE SIGNS

(49) **20** tool unit **21** press **22** press frame **23** holder **24** frame guide device **25** movement guide device **26** guide projection **27** ram **28** ram drive **31** tool carrier **32** movement recess **33** top groove flank **34** bottom groove flank **38** sensor device **39** light transmitter **40** light receiver **41** through-hole **45** first tool **46** second tool **47** tool guide device **48** guide column **49** guide bushing **50** transverse support **51** first pressure piece **52** first embossing stamp **53** second pressure piece **54** second embossing stamp **57** transport part **58** slide surface **59** round blank **62** ejector device **63** ejector **64** ring **68** coupling device **69** tool coupling surface **70** stop surface **71** ram surface **72** coupling part **73** actuator unit **74** biasing device **75** counter coupling part **76** coupling surface **77** counter coupling surface **82** positioning device **86** positioning body **84** positioning drive **85** positioning recess **90** clamping device **91** clamping body **92** clamping drive **93** disk spring stack **94** piston **95** cylinder I forming position II ejecting position III alignment position IV clamping position A tool axis H stroke direction R back-stroke direction

Claims

1. A press (**21**) for forming a round blank (**59**) comprising: a holder (**23**) for an exchangeable tool unit (**20**) that comprises a first tool (**45**) and a second tool (**46**) that is movably supported in a stroke direction (H) toward the first tool (**45**) and in a back-stroke direction (R) away from the first tool (**45**) along the tool axis (A), a positioning device (**82**) that comprises at least one positioning body (**83**) that cooperates with one assigned positioning recess (**85**) on the tool unit (**20**) respectively in order to position and/or align the tool unit (**20**) relative to the press (**21**), the positioning recess (**85**) has a form tapering from an open end toward the other end of the positioning recess (**85**) and the at least one positioning body (**83**) tapers in the backstroke direction R toward its free end, and, a clamping device (**90**) that is configured to clamp the tool unit (**20**) arranged in the holder (**23**) after the positioning device (**82**) has positioned and/or orientated the tool unit (**20**) relative to the press (**21**).
2. The press according to claim 1, further comprising a coupling device (**68**) is provided that is configured to couple the second tool (**46**) with a ram (**27**) of the press (**21**), such that the movement of the ram (**27**) is transferred to the second tool (**46**) at least in a movement area of the ram (**27**).
3. The press according to claim 2, further comprising the coupling device (**68**) is configured to urge the second tool (**46**) in the back-stroke direction (R) with a coupling force toward the ram (**27**).
4. The press according to claim 3, further comprising a stop surface (**70**) is provided against which the second tool (**46**) is urged due to the clamping force in a rest position.
5. The press according to claim 2, further comprising the coupling device (**68**) is configured to allow transverse movements of the ram (**27**) transverse to the tool axis (A) and/or tilting

movements of the ram (27) relative to the tool axis (A).

6. The press according to claim 1, further comprising a frame guide device (24) is provided in the holder (23) that cooperates with a movement guide device (25) provided on the tool unit (20).

7. The press according to claim 6, wherein a clearance is provided between the frame guide device (24) and the movement guide device (25) configured for the possibility of relative movement between the tool unit (20) and the press frame (22).

8. The press according to claim 1, further comprising a positioning drive (84) is provided and configured to move the positioning bodies (83) in stroke direction (H) and backstroke direction (R) and can be moved into an alignment position (III), wherein the positioning bodies (83) project into the respective assigned positioning recess (85) and abuts against the tool carrier (31) inside the respective assigned positioning recess (85).

9. The press according to claim 1, further comprising a sensor device (38) is provided and configured to detect that the tool unit (20) is completely and correctly arranged in the holder (23).

10. The press according to claim 1, wherein the exchangeable tool unit (20) comprises: a tool carrier (31) that is configured to be exchangeably connected with the press (21), the first tool (45) and the second tool (46) are arranged in alignment with the tool axis (A) on the tool carrier (31), wherein the second tool (46) is movably arranged in a guided manner on the tool carrier (31) along the tool axis (A) by means of a tool guide device (47) such that the second tool (46) is movable in the stroke direction (H) toward the first tool (45) and in the back-stroke direction (R) away from the first tool (45), wherein the second tool (46) is configured to be movably coupled with a ram (27) of the press (21).

11. The press according to claim 10, wherein the tool guide device (47) is configured to support transverse movements of the second tool (46) transverse to the tool axis (A) and/or tilting movements of the second tool (46) relative to the tool axis (A).

12. The press according to claim 10, wherein the tool guide device (47) comprises at least one guide column (48) that is movably supported in an assigned guide bushing (49) in the stroke direction (H) and the back-stroke direction (R).

13. The press according to claim 12, wherein the second tool (46) is movably coupled with the at least one guide column (48) and that the at least one guide bushing (49) is arranged on the tool carrier (31).

14. The press according to claim 10, wherein the second tool (46) comprises a tool coupling surface (69) that is configured to be in abutment against a ram surface (71) of a ram (27) of the press (21) during movement of the second tool (46) in stroke direction (H).

15. The press according to claim 10, wherein the first tool (45) is movable relative to the tool carrier (31) between a forming position (I) and an ejecting position (II).

16. The press according to claim 15, wherein in the forming position (I) the first tool (45) is immovably supported in stroke direction (H) on the tool carrier (31) and/or the press (21).

17. The press according to claim 10, further comprising a transport part (57) having a sliding surface (58) for sliding transport of round blanks (59) into the tool unit (20) and out of the tool unit (20) is provided on the tool carrier (31).

18. The press according to claim 10, further comprising a movement guide device (25) is provided on the tool carrier (31) that is configured to cooperate with a frame guide device (24) on a press frame (22) of the press (21) in order to move the tool unit (20) in an insertion direction into the press (21) or out of the press (21).

19. The press according to claim 10, wherein the one positioning recess (85) is provided on the tool carrier (31) that is configured to cooperate with the at least one positioning body (83) of the press (21) in order to define a relative position and/or relative orientation between the tool unit (20) and the press (21).

20. A method of operating the press for forming the round blank according to claim 1, said method comprising the following steps: Inserting the tool unit (20) in the holder (23) of the press (21),

Positioning and/or aligning the tool unit (20) relative to a press frame (22) by means of the positioning device (82) in that the at least one positioning body (83) of the positioning device (82) is projecting in the associated positioning recess (85) on the tool unit (20), the positioning recess (85) has a form tapering from the open end toward the other end of the positioning recess (85) and the at least one positioning body (83) tapers in the backstroke direction R toward its free end, Clamping the tool unit (20) on the press frame (22) by means of a clamping device (90), and Retracting the at least one positioning body (83) out of the positioning recess (85) on the tool unit (20).

21. The method according to claim 20, wherein the tool unit (20) comprises the first tool (45) and the second tool (46) and that the second tool (46) is movably coupled by means of a coupling device (68) with a ram (27) of the press (21) subsequent to positioning and/or aligning the tool unit (20) relative to the press frame (22).
